

Dylan Reiser

Mechanical Engineering Portfolio

SELECTED WORK - 2024-2026

I'm drawn to problems nobody has solved yet. My work usually starts with a question I'm curious about, one I can't answer from a desk. So, I go find out: thirty-plus interviews with construction workers before a single part was modeled, fit checks in an actual bus before a mount was approved. Then I learn it, sketch it, build it, break it, and try again. The pieces I'm proudest of came from working beside people who knew things I didn't, across electrical, industrial design, and manufacturing, helping me to learn more along the way.

PROFILE

School	Georgia Institute of Technology B.S. Mechanical Engineering, Minor in Technology & Business · GPA 3.85
Abroad	Georgia Tech-Europe, Metz, France · 2026
Now	· Peer Instructor, GT Invention Studio · Alumni Connections Chair, Student Alumni Association · Alumni Relations, Alpha Tau Omega
Tools	SolidWorks · FEA · Flow Simulation · GD&T PCB design · Embedded C++ · MATLAB · CNC Additive Manufacturing

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- 01 **HelmSight — Hard Hat Camera**
Enclosure, PCB, and firmware - GT CreateX
- 02 **Solar Thermal Sand Battery**
Flow routing and valve design - GHP
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In-vehicle mounting hardware - Internship



FIG 01 — ME, WEARING THE HELMSIGHT DEVICE

Final, fully functional prototype for the HelmSight communication and safety device. Attached to the brimmed hard-hat via a discrete clamp mount.

01 HelmSight

Hard hat camera and jobsite safety device

GT CREATEX IDEA2PROTOTYPE - ONE SEMESTER - ENCLOSURE - PCB - FIRMWARE - ON-SITE TESTING

Thirty-plus interviews with construction workers, foremen, and project managers landed on the same bottleneck: communication up the chain of command. **HelmSight slots in or clamps onto a standard hard hat and captures a photo, a voice note, and a GPS fix from one button press, enabling quick, hands-free documentation and communication** that can be easily shared. I carried it from that first interview to a working prototype, designing, manufacturing, and validating the enclosure, the electronics, and the firmware.

- **Designed the enclosure and brim mount in SolidWorks**, applying DFM and DFA to reach a lightweight unit that clamps to existing hard hats with no modification to the helmet.
- **Ran static FEA on the accessory-slot mount** and revised the geometry to move stress off the insert corner, then confirmed the change on printed parts.
- **Laid out a custom two-layer PCB** carrying an ESP32-S3 camera module, PDM microphone, GPS, buzzer, and lithium charging, then hand-assembled and tested the first boards.
- **Wrote the embedded C++ firmware**: synchronized photo, audio, and GPS capture to SD, a one-button interface with short, double, and long press, and a Wi-Fi access point serving a mobile file browser with live preview.
- **Held the bill of materials near \$71 per unit** and validated fit, clearance, and wearability across four printed enclosure generations.

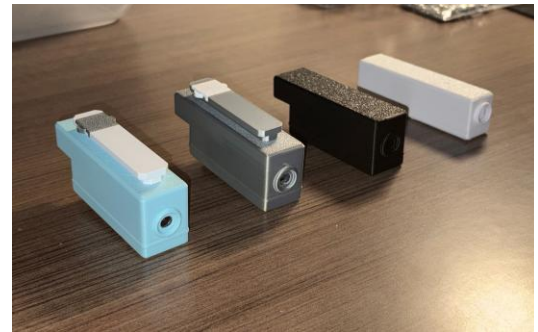


FIG 02 – ENCLOSURE ITERATIONS

Four printed generations, oldest at left. Each round adjusted wall thickness, lens standoff, and lid retention after fit checks on a real hard hat.

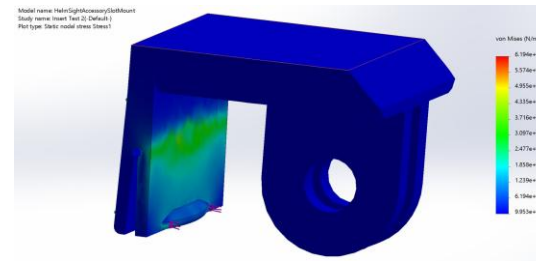


FIG 03 – MOUNT STRESS STUDY

Static nodal stress on the accessory slot-mount. The plot drove a revision that pulled load off the insert corner, validated in real-world wear testing. Testing was repeated on clamp-mount too.

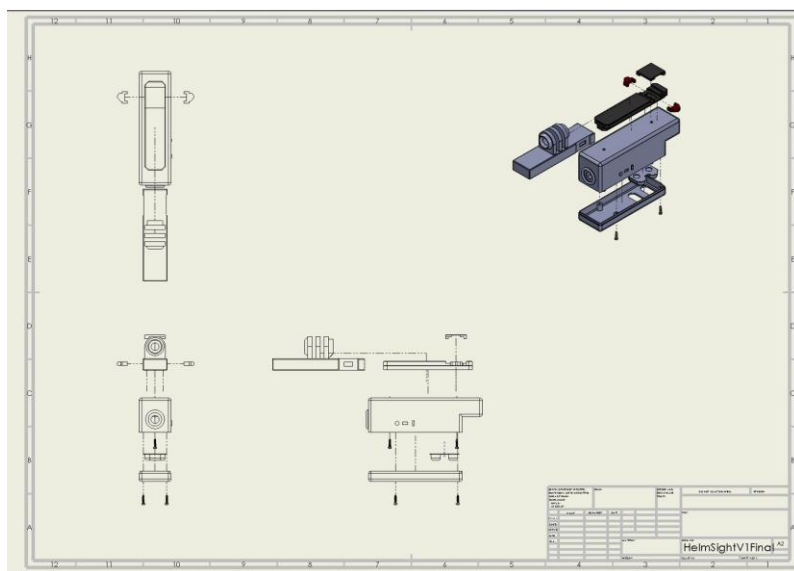


FIG 05 – V1 ASSEMBLY DRAWING

Exploded views of the final enclosure: electronics enclosure, button insert, rail mount with push-pins, and fasteners.

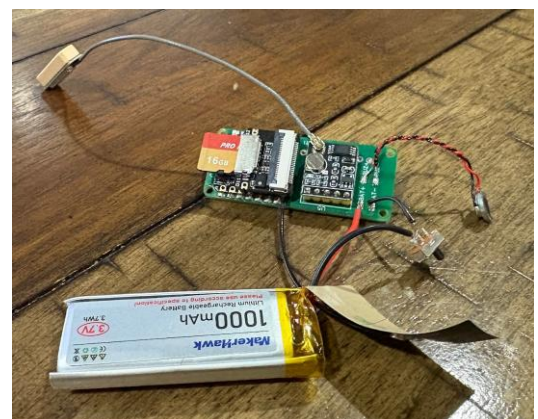


FIG 04 – FIRST ASSEMBLED BOARD

Bring-up with the 1000 mAh cell, external antenna, and microSD seated. Testing here found a buzzer-to-SD pin conflict, fixed by reordering peripheral initialization.

02 Solar Thermal Sand Battery

Sustainable energy storage — water loop and valve design

GOVERNOR'S HONORS PROGRAM - ENGINEERING DESIGN - 20-PERSON TEAM - 1 OF 600 FROM 4,200+ NOMINEES

More than anything, this project, and the Governor's Honors Program as a whole, taught me how to effectively collaborate and communicate in a workspace environment. **Working within a 20-person team, I lead a 5-person sub-team to design, develop, and test the fluid systems that extract thermal energy from the battery for sustainable energy storage.** We performed flow and thermal simulations to determine the specs required for the system pump and cycle time to heat the water to desired temperature. We used Monay.com work management platform, worked many late nights, and had fun while we were doing it.

- **Modeled the water routing in SolidWorks and ran Flow Simulation** on the 3D printed connectors feeding the thermal coil, checking velocity and pressure through each stepped transition.
- **Designed, printed, and tested multiple valve geometries**, iterating on sealing and flow until the assembly held under pump pressure.
- **Integrated the water loop** with the insulated sand enclosure and the team's control and sensing subsystems, then helped wire and continuity-check the electronics ahead of test runs.
- **Documented the system in Bluebeam** and presented the final design to a panel of industry engineers and managers.

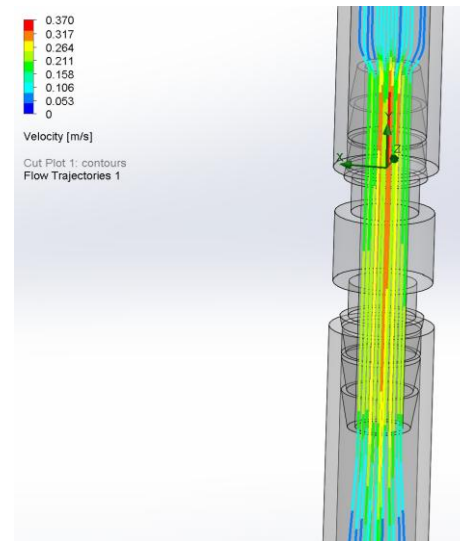


FIG 06 – CONNECTOR FLOW STUDY

Flow Simulation trajectories through the printed connector stack, colored by velocity. Used to check how much fittings would affect flow rate and pressure.



FIG 07 – FULL TEST ASSEMBLY

The copper thermal coil sits inside the insulated sand enclosure, plumbed to the reservoir and pump stand at right. When the pump cycles, the coil pulls stored heat out of the sand and delivers hot water. We were testing continuity of all electrical systems and sensors in the photo.

03 Passio Technologies

In-vehicle mounting hardware and manufacturing documentation

SUMMER ENGINEERING INTERN - APRIL-AUGUST 2025 - SANDY SPRINGS, GA

Passio builds passenger counting and transit tracking systems that run on public bus fleets. My work focused on two things: **the mechanical design of the hardware that goes inside the bus, and the wiring documentation that overseas manufacturing builds from.** Both had to respond to vehicles and systems already in service, which meant designing around existing device geometry, cable routing, and thinking about real-world installation and technician access.

- **Designed mounting surfaces for in-bus sensors and tablets in SolidWorks**, running FEA for static and vibration loading and printing iterations for fit checks.
- **Created and optimized wiring harness diagrams** for overseas manufacturing of bus sensors and tablets deployed across more than 12,000 buses.
- **Carried designs through to installed parts**, closing the loop between a printed fit check on the bench and a mount that a technician can service in the field.

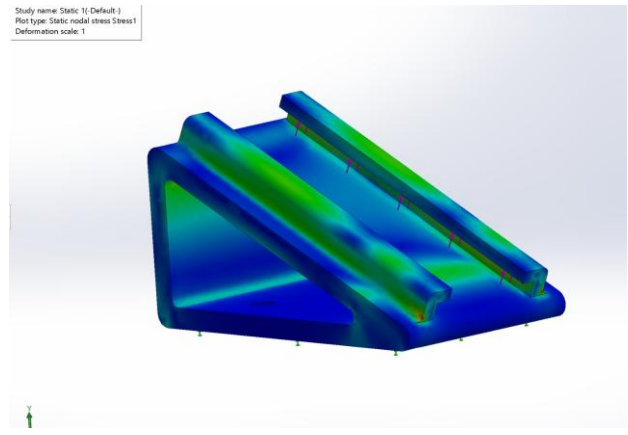


FIG 08 – RAIL MOUNT LOAD STUDY

Static nodal stress on a test rail mount for a passenger counting camera, loaded along the rail contacts and fixed at the base fastener points.

OTHER EXPERIENCE and PROJECTS

GT Invention Studio — Peer Instructor. Certify and train incoming instructors on 30+ metalworking, woodworking, and additive tools, and support student teams through CAD review, fabrication, and troubleshooting.

Solid model rocket motor. Founded my school's first engineering club and led a team of four through propellant and casing iterations to 2 lbs peak thrust and 170 s specific impulse, then built and tested the parachute ejection charges from scratch.

Student Alumni Association — Alumni Connections Chair. Run the outreach database and 10+ events a semester connecting 300+ students with alumni.

SKILLS and CERTIFICATIONS

Design & analysis. SolidWorks (CAD, FEA, Flow Simulation), GD&T, PCB design, embedded C++, MATLAB, Bluebeam.

Manufacturing. CNC machining, additive manufacturing, design for manufacturability and assembly, shop safety instruction.

Certifications. CSWA, CPR/AED, Water Rescue.

Coursework abroad. 10 credits of mechanical engineering at Georgia Tech—Europe, Metz, France.

Looking for a summer 2027 internship in mechanical systems, product development, or design prototyping.